

Swiggy

Swiggy Food Intelligence

An END -to- END Data Pipeline & Analysis project with Streamlit Dashboard and Integrated LLM & RAG chatbot

SNOWFLAKE – DBT – PYTHON – STREAMLIT – LLM & RAG

Swiggy Food Delivery Analytics

Snowflake · dbt · ELT · Python analytics · quarterly city revenue forecasting · Streamlit · Rag chatbot

> *Data integrity note: This is a learning and portfolio project built with synthetic order data. The figures, patterns, model scores, and forecast are produced from synthetic data. They do not describe Swiggy's real customers, restaurants, revenue, service quality, or business performance and must not be treated as real-world evidence.*

1. Introduction

This project demonstrates an end-to-end analytics workflow for a food-delivery business. It begins with CSV datasets, loads them into Snowflake, transforms them with dbt into analysis-ready models, explores them in Python, builds a quarterly revenue forecast for each city, and presents business views in a Streamlit application with an LLM-assisted question interface.

The main modeling question is: How can a city's next-quarter revenue be estimated from its historical revenue and order patterns? The notebook compares a learned model with simple historical baselines and then uses the selected model to produce a 2026 Q3 forecast. The dashboard lets a user explore city, restaurant, customer, delivery, and forecast views.

The project is intended to demonstrate tools and analytical reasoning. Because the order records are synthetic and the available history is limited, the results do not validate a real operational decision or prove that the same model would work on production data.

2. Related concepts

2.1 ELT and ETL

Both patterns move source data into a form that can be analyzed. ETL means Extract, Transform, Load: data is transformed before it enters the target warehouse. ELT means Extract, Load, Transform: source data is loaded first, and transformations are then run inside the warehouse. This project follows ELT: local CSV files are uploaded to a Snowflake stage, copied into RAW tables, and then transformed by dbt SQL models in Snowflake.

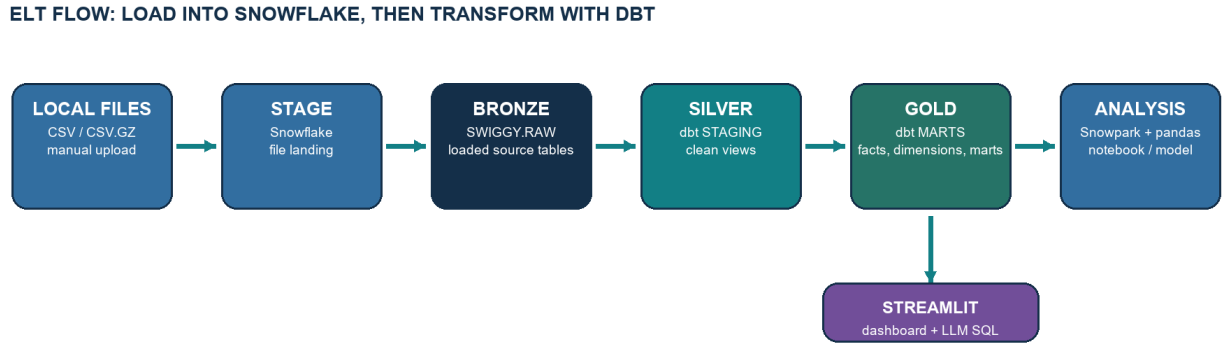


Figure: project ELT and medallion architecture

Snowflake separates managed data storage from virtual warehouse compute. In this project, the Snowflake warehouse runs the SQL statements issued by loading commands, dbt, the notebook connection, and the Streamlit app. A stage stores incoming files; a RAW table stores queryable rows after the COPY INTO operation.

2.2 Bronze, Silver, and Gold: the medallion pattern

The medallion architecture is a data-refinement pattern often described with Bronze, Silver, and Gold layers. The labels describe increasing readiness for use; they are not special Snowflake object types. This project maps the pattern to its actual schemas and dbt model groups:

Conceptual layer	Project implementation	Purpose	Examples
Bronze	SWIGGY.RAW	Keep source-shaped data available after loading, so the source can be checked and traced.	RAW.ORDERS, RAW.RESTAURANTS, RAW.ORDER_ITEMS
Silver	SWIGGY.STAGING dbt views	Standardize names and types, clean source values, and create reusable records.	STG_ORDERS, STG_RESTAURANTS, STG_ORDER_ITEMS
Gold	SWIGGY.MARTS dbt tables	Organize trusted analysis outputs into facts, dimensions, and focused summary marts.	FACT_ORDERS, DIM_DATE, MART_DAILY_CITY_REVENUE

So, in this project, Bronze corresponds to RAW, Silver to STAGING, and Gold to MARTS. There are no schemas literally named BRONZE, SILVER, or GOLD.

2.3 How dbt fits into the ELT workflow

dbt is the SQL transformation and analytics-modeling layer. Each SQL model describes a SELECT query and how its result should be materialized in the warehouse. ref() connects a model to another dbt model. Those references let dbt build a dependency graph, or DAG, and determine the required build order. In this project, dbt staging models are views and mart models are tables, as configured in dbt_project.yml.

The project also uses YAML data tests. Examples include uniqueness and non-null checks for identifiers, allowed values for order status, and a relationship check from orders to customers. A data test checks an explicit assumption; it does not prove that every business rule or source value is correct.

dbt transforms data already in Snowflake. It is not the command used to upload local files to a stage, and it is not the notebook or dashboard. The project runs its loading and dbt commands manually; it does not use Airflow or a scheduled production orchestrator.

2.4 Gold facts, dimensions, marts, and star-schema modeling

A fact table records measurable business events at a declared grain, the meaning of one row. A dimension table holds descriptive context for grouping and filtering facts. A star schema arranges facts with their dimensions so common analytical questions can be answered with clear joins and aggregations. The Gold layer in this project is star-schema style, with two fact tables at different grains and several supporting marts.

Gold model	One row represents	Example use
FACT_ORDERS	One order	Order revenue, order counts, delivery time, customer ratings
FACT_ORDER_ITEMS	One food item line within an order	Item quantity, price, line amount, food-level analysis
DIM_RESTAURANTS	One restaurant	Group by restaurant, city, cuisine, rating, or listed cost
DIM_CUSTOMER	One customer	Group by age and customer profile attributes
DIM_FOOD	One food item	Group by food name or vegetarian classification
DIM_DATE	One calendar day	Group by date, month, quarter, weekday, or weekend
MART_DAILY_CITY_REVENUE	One city per order date	City revenue and order trends
MART_DELIVERY_SLA	One city per order hour	Median and 90th percentile delivery time
MART_RESTAURANT_PERFORMANCE	One restaurant	Restaurant order, revenue, rating, and delivery summaries

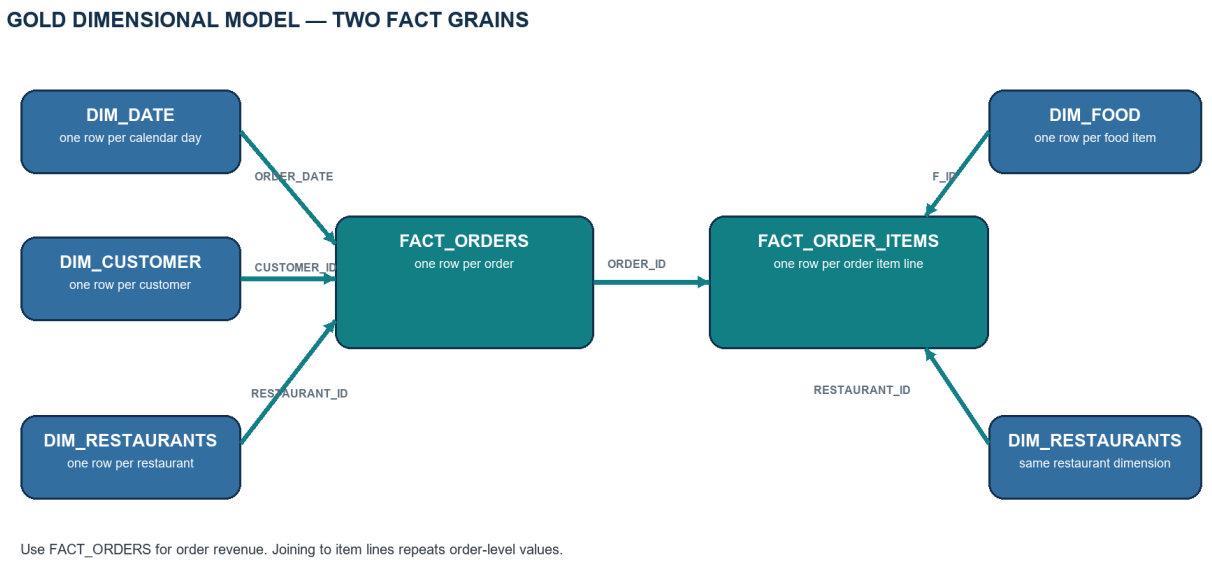


Figure: Gold fact and dimension relationships

The facts use source/business identifiers for joins. The two fact grains matter: an order with multiple item lines occurs once in FACT_ORDERS and multiple times in FACT_ORDER_ITEMS. Joining them and then summing order-level SALES_AMOUNT would repeat that amount for each item line. Use the order fact for order-level revenue and the item fact for item-level measures, or aggregate carefully before combining them.

The summary marts are Gold products for common business questions. They reduce repeated query work for views such as daily city revenue, restaurant performance, and delivery-time percentiles. They are separate from the detailed facts and dimensions.

2.5 Python, Snowpark, and pandas

The notebook creates a Snowpark session to connect to Snowflake and select the Gold tables. It converts selected results into pandas DataFrames for local notebook work. Snowpark provides the Snowflake connection and query/DataFrame interface; pandas, NumPy, and plotting libraries support the analysis after data is brought into the notebook. This is a Python analysis consumer of the dbt Gold layer, not another dbt layer.

2.6 Analysis concepts used

- Profiling and data-quality checks summarize row counts, columns, data types, missing values, duplicates, and value ranges before interpreting metrics.
- Univariate analysis studies one variable at a time, such as revenue distributions or customer-age frequencies.
- Bivariate and multivariate analysis compares measures across categories and combinations of variables, such as revenue by city, cuisine, customer segment, and month.
- Time-series analysis preserves chronological order to inspect revenue and order patterns over months and quarters. Quarter revenue is also normalized by quarter length where appropriate, since calendar quarters have different numbers of days.
- Root-cause analysis in this notebook is exploratory and descriptive: it compares where and when revenue changes and which segments differ. These observational comparisons do not establish causality.
- Feature engineering creates predictors using information available before the forecast quarter. This reduces leakage, where a model accidentally receives information about the outcome it is supposed to predict.
- Time-ordered model evaluation uses earlier data for training, a later quarter for validation, and a still later quarter for a holdout test. WMAPE summarizes total absolute forecast error relative to total actual revenue. Baselines are necessary to check whether a learned model improves on a simple rule.

2.7 Streamlit and LLM-assisted questions

Streamlit provides the interactive dashboard interface. The app reads Snowflake Gold models to display KPI cards, filtered summaries, charts, and forecast results. Its Ask Your Data feature sends a natural-language question to an LLM, requests a constrained SQL response, checks that it is a single read-only SELECT, runs it against allowed Gold tables, and formats the result as text, a table, or a chart.

The current app interface calls this feature a RAG chatbot, but the inspected implementation is LLM-assisted text-to-SQL. It does not contain a document embedding and vector-search retrieval pipeline. A future RAG implementation would need a retrieval source, such as business definitions or project documentation, and a retrieval step before generation; that is not part of the current code.

2.8 Official concept references

- Snowflake, Key concepts and architecture (https://docs.snowflake.com/en/user-guide/intro-key-concepts).
- Snowflake, Bronze and Silver/Gold layer responsibilities in an Openflow example (https://docs.snowflake.com/en/en/user-guide/data-integration/openflow/about-spcs).
- dbt Labs, Quickstart for dbt and Snowflake (https://docs.getdbt.com/guides/snowflake).
- dbt Labs, The ref function and model dependencies (https://docs.getdbt.com/reference/dbt-jinja-functions/ref).
- dbt Labs, Data tests (https://docs.getdbt.com/docs/build/data-tests).

3. Project preparation

3.1 Data and project setup

The project folder contains CSV files for restaurants, users, food, menu, order items, and reviews. The original set did not include an order dataset, so an order dataset was prepared for the learning project. The order records are synthetic. The project directory does not include a reproducible generator script or seed for recreating that file, so the synthetic data-generation recipe cannot be independently repeated from the repository alone.

The local project was organized with a data folder, a dbt project (swiggy_dbt), an analysis notebook, a Streamlit application, and documentation. Snowflake was configured with the SWIGGY database, SWIGGY_WH warehouse, RAW/STAGING/MARTS schemas, and the local internal stage. dbt was configured with a Snowflake profile and model materializations for staging views and mart tables.

3.2 Preparation sequence

1. Inspect the available CSV files and identify the missing order dataset needed to connect orders, users, restaurants, and line items.

2. Prepare the synthetic order CSV for the analysis workflow and keep its synthetic nature visible in the project documentation.

3. Create Snowflake file-format and internal-stage objects, then upload the compressed CSV files to SWIGGY_LOCAL_STAGE.

4. Define RAW tables with source-aligned columns and types, and load staged data with COPY INTO.

5. Configure the dbt Snowflake profile and project model paths and schemas.

6. Create notebook and application environments with the Snowflake and analysis libraries needed for Python work.

7. Keep credentials in local environment configuration or secrets and outside the public repository.

4. Implementation

4.1 Snowflake loading: local files to RAW

The local CSV files were uploaded to the Snowflake internal stage. COPY INTO then parsed those staged files using the configured CSV file format and loaded rows into the RAW tables. This is the load part of ELT. The stage is the file landing area; RAW is the database layer holding loaded records.

4.2 Silver: dbt staging views

One staging model was created per source entity. The models standardize names and types, handle source formatting, and select records into consistent views. This is the reusable cleanup layer between source-

shaped RAW data and business-oriented Gold models. YAML tests cover selected key assumptions. The staging run and dbt tests completed successfully in the project workflow.

4.3 Gold: dimensions, facts, and marts

The Gold layer contains dimensions for restaurants, customers, food, and dates; incremental facts for orders and order items; and aggregate marts for daily city revenue, delivery SLA, and restaurant performance. Each model uses the staging layer as its input through dbt references. The marts are materialized as tables, while staging is materialized as views.

4.4 Notebook analysis

The notebook opens a Snowpark connection to Snowflake and loads Gold tables for analysis. It checks DataFrame shape, types, missingness, duplicates, and distributions before building analysis views. The saved notebook contains distribution analysis, category-based comparisons, monthly and quarterly time series, descriptive revenue-change comparisons, feature engineering, model evaluation, and charts.

Important data-quality findings affect interpretation: CUSTOMER_RATING is missing for 9,704,459 of 10,000,000 orders (97.04%). Every row in the Gold FACT_ORDERS output is marked Delivered, so cancellations are not available for analysis in that output. The notebook also identifies an extreme restaurant cost value and caps it for restaurant analysis; that cost adjustment is not a feature in the city-quarter revenue model.

4.5 Quarterly city revenue model

Daily city revenue was too noisy for the selected forecast task, so the model dataset is built at one row per city per quarter. The prediction target is next-quarter city revenue, using past revenue and lagged order-pattern features. Features include historical revenue ratios, city size, history length, quarter, average order value, customers per order, items per order, discount share, and UPI share. The current quarter's outcome is excluded from its predictors.

The chronological evaluation uses 2026 Q1 for validation and 2026 Q2 as a holdout test. The selected LightGBM model was selected by validation WMAPE; the holdout is kept separate as a final check. A historical-average baseline is compared with it. The Streamlit forecast function uses the saved model artifact and the notebook's feature and scaling contract to calculate the forecast.

4.6 Streamlit application

The app provides Executive Overview, City & Restaurant Analysis, Customer & Delivery Analysis, Revenue Forecasting, and Ask Your Data views. It reads curated Gold tables from Snowflake, applies date and city filters, visualizes metrics, and uses a saved city revenue model artifact for forecasting. The LLM-assisted question workflow is described in Section 2.7.

5. Results and demo

5.1 Data and analysis summary

Measure	Saved result
Orders in Gold FACT_ORDERS	10,000,000
Order period	Jan 2024 – Jun 2026
Cities in quarterly forecast panel	552
Orders missing customer rating	9,704,459 (97.04%)
City-quarter training panel rows	4,416
Forecast horizon shown	2026 Q3

Bangalore leads city revenue in the saved analysis with INR 608.1M (12.13% share); the top five cities account for about 44.5%. UPI is used for about 55% of orders. These figures describe the synthetic project data only.

5.2 Holdout model results: 2026 Q2

Method	MAE (INR)	RMSE (INR)	WMAPE	Bias
Last quarter	17,507.68	37,461.57	1.93%	1.0016
Average of past quarters	12,348.12	25,233.26	1.36%	1.0009
Ridge	12,285.99	25,145.05	1.36%	0.9991
LightGBM, chosen on validation	12,645.87	26,348.52	1.40%	1.0019

LightGBM had the best validation WMAPE (1.37%) and was selected by the notebook's rule. On the later 2026 Q2 holdout, LightGBM did not beat the historical-average baseline. The model's improvement over a simple baseline is therefore not demonstrated by this holdout result.

5.3 Next-quarter forecast

Forecast summary	Saved result
Total 2026 Q3 forecast across 552 cities	INR 506.25M
2026 Q2 daily pace scaled to 92 days	INR 505.49M
Difference from that reference	About +0.15%
Largest top-ten city watch	Ahmedabad: Growth watch (+1.47%); nine others Stable

The watch flags are prompts to review when new quarters arrive. The forecast is an output on synthetic data, not a real revenue prediction.

5.4 Result screenshots and demo

App screenshots will be inserted by the project owner. Suggested screenshots: Executive Overview, City & Restaurant Analysis, Revenue Forecasting, and Ask Your Data. Keep the screenshot captions tied to the synthetic dataset and avoid presenting synthetic values as real business results.

Executive Overview screenshot goes here	Revenue Forecasting screenshot goes here	Ask Your Data screenshot goes here

Screenshot to add	Suggested caption
Executive Overview	Main KPIs and selected time/city filters (synthetic project data)
City & Restaurant Analysis	City revenue and restaurant comparison views
Revenue Forecasting	Actual quarters and 2026 Q3 city-revenue forecast

Ask Your Data	Example natural-language question, generated read-only SQL, and resulting chart/table

5.5 Limitations and interpretation

- All order data is synthetic; the figures do not describe Swiggy's real operations.
- Ratings are present for only about 3% of orders, so they do not represent the full customer experience.
- Every Gold order is Delivered; cancellations cannot be analyzed from the current fact output.
- The root-cause section describes patterns and associations; it does not prove causal drivers.
- The holdout contains one quarter, so model performance needs more real, later-period data before any operational use.
- The current Ask Your Data feature is text-to-SQL; it is not a document RAG system.
- The app needs an active Snowflake account and configured LLM credentials for live queries. Credentials should never be committed to GitHub.

5.6 Notebook figures

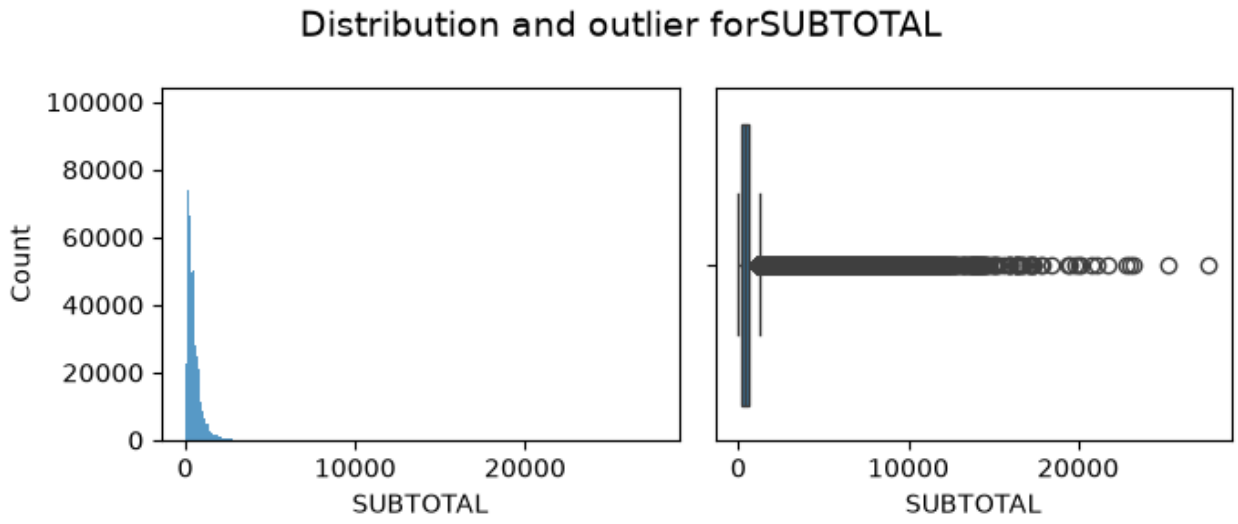
The Word report includes every saved notebook chart output. These figures were extracted from the saved notebook; they were not rerun from Snowflake while preparing this report.

5.7 Analysis chart gallery

The following gallery reproduces the saved figures from the analysis notebook. The chart outputs are included as they were saved; they have not been recalculated for this report.

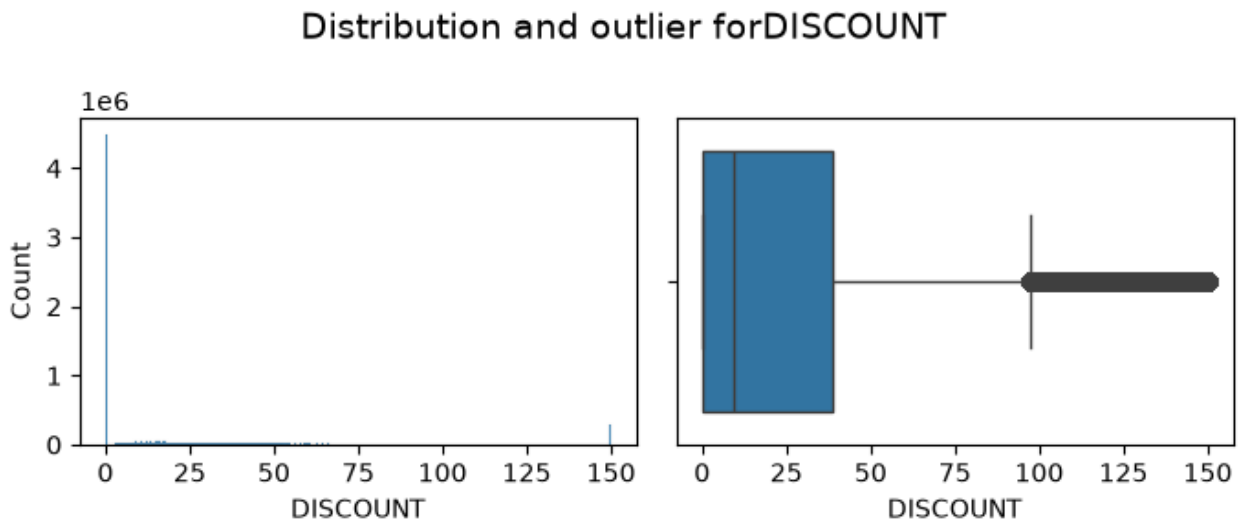
Distribution analysis

Order financial measures: subtotal



Order subtotal distribution and outliers

Order financial measures: discount



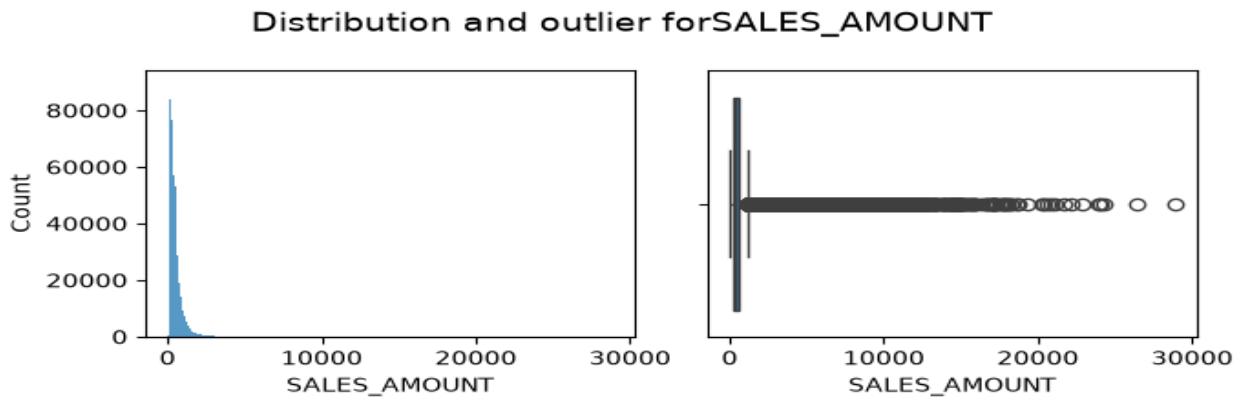
Order discount distribution and outliers

Order financial measures: delivery fee



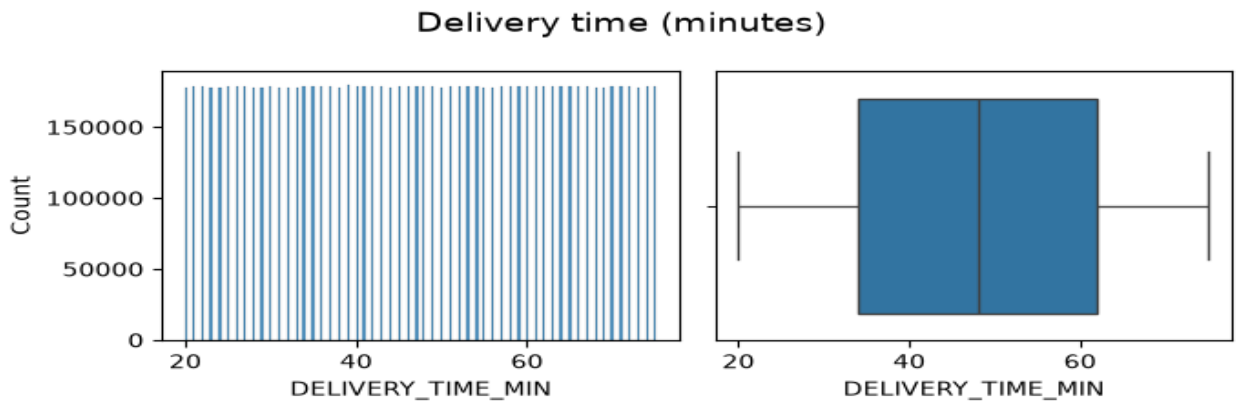
Delivery fee distribution and outliers

Order financial measures: sales amount



Sales amount distribution and outliers

Order experience and basket measures: delivery time



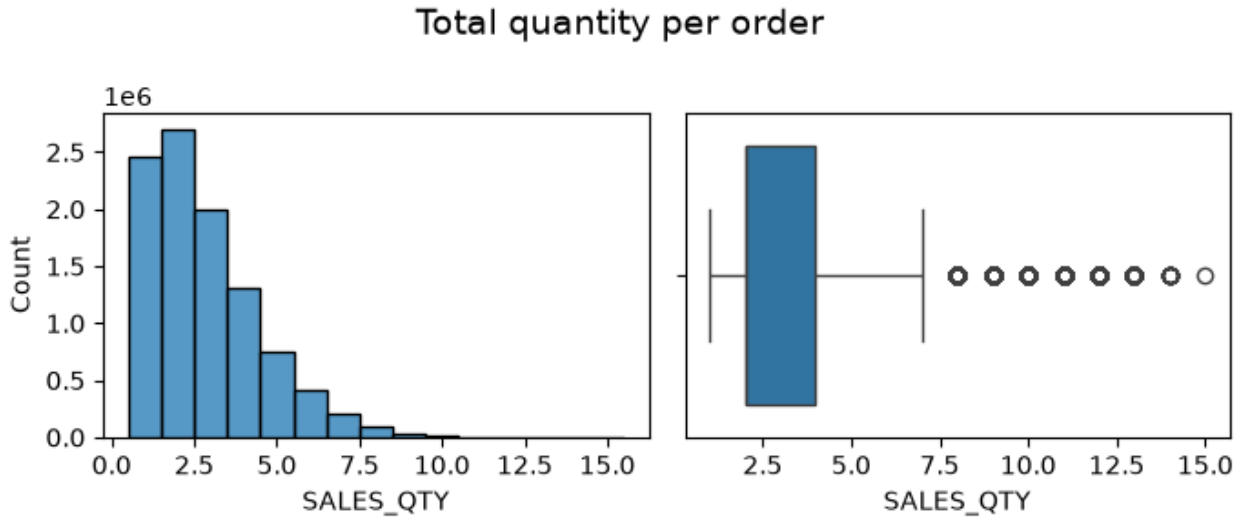
Delivery time distribution

Order experience and basket measures: items per order



Items per order distribution

Order experience and basket measures: sales quantity



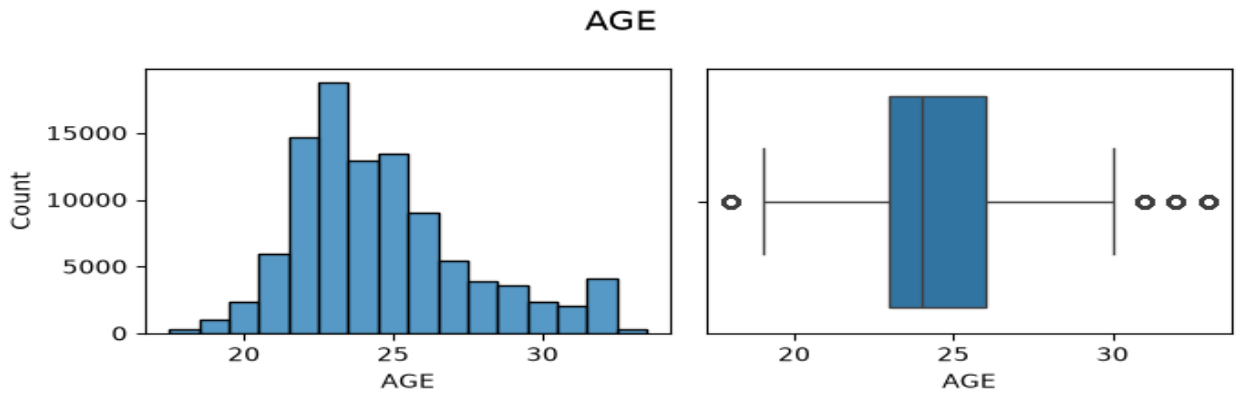
Sales quantity per order distribution

Order experience and basket measures: customer rating



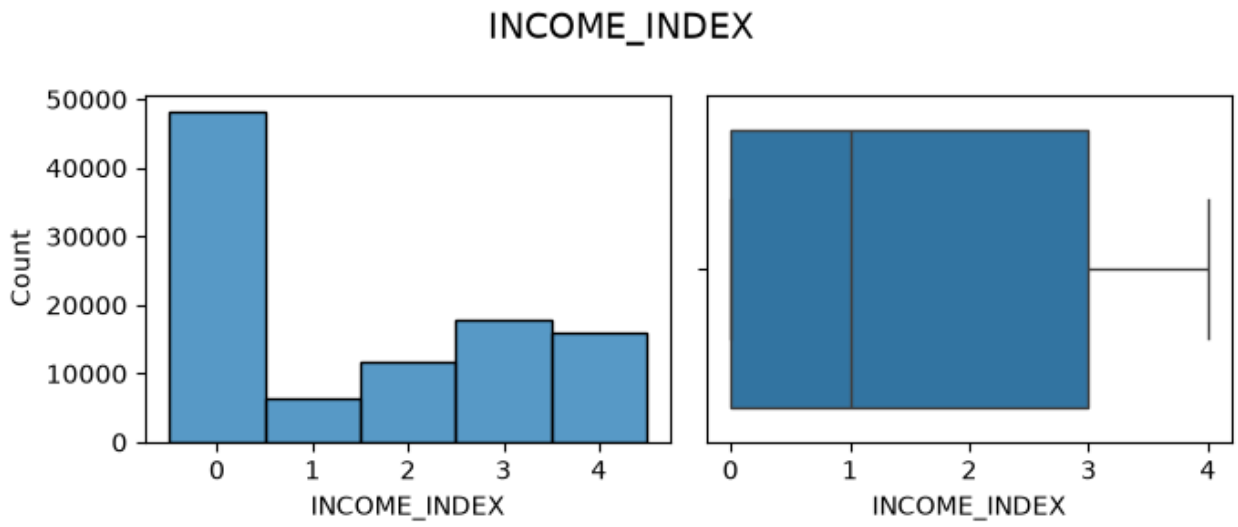
Customer rating distribution

Customer profile: age



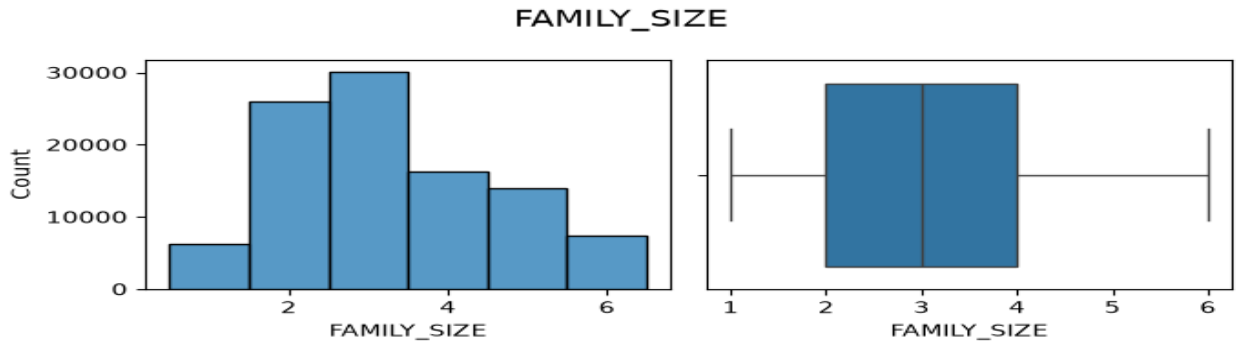
Customer age distribution

Customer profile: income and family measures



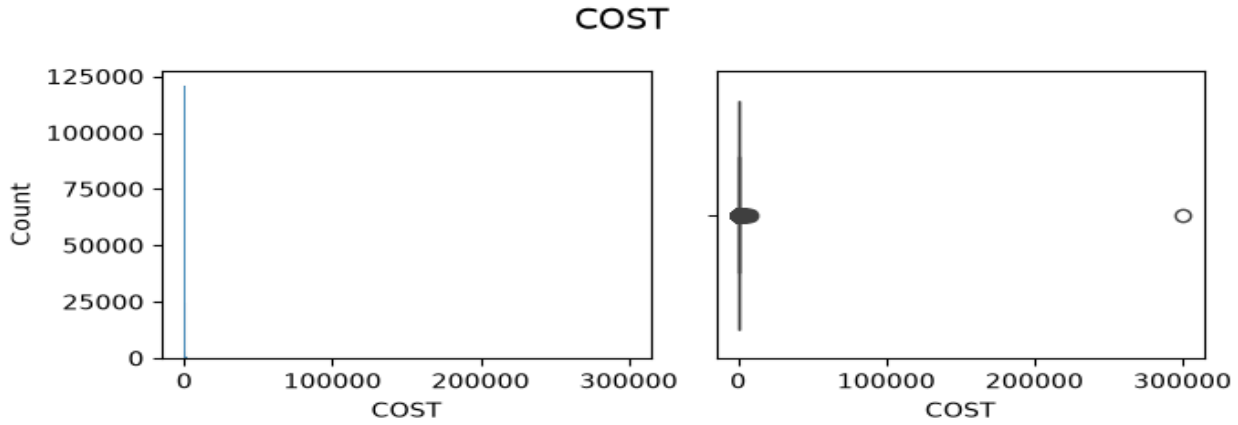
Customer income and family profile distribution

Customer profile: categorical attributes



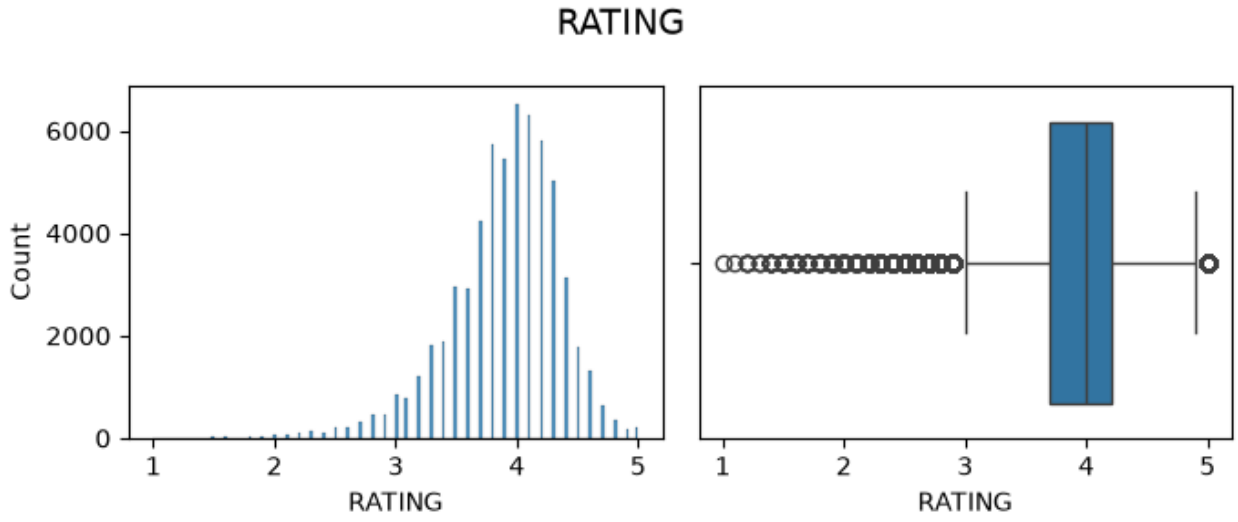
Customer category distribution

Restaurant profile: cost



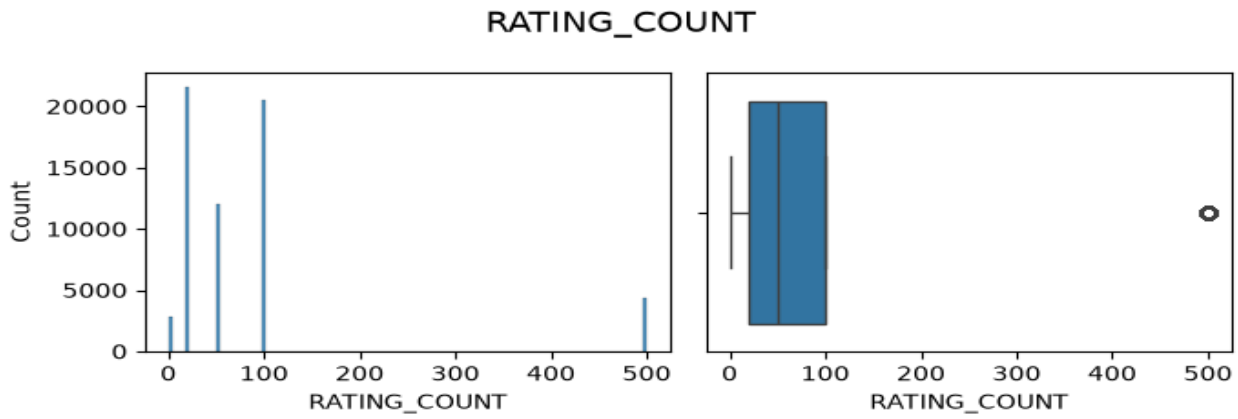
Restaurant cost distribution

Restaurant profile: rating



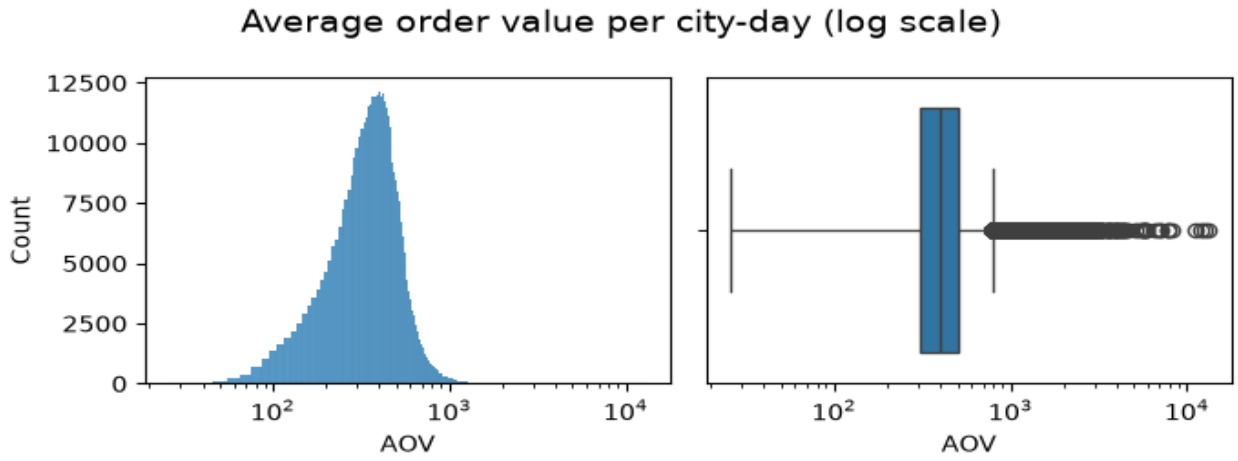
Restaurant rating distribution

Restaurant profile: rating count



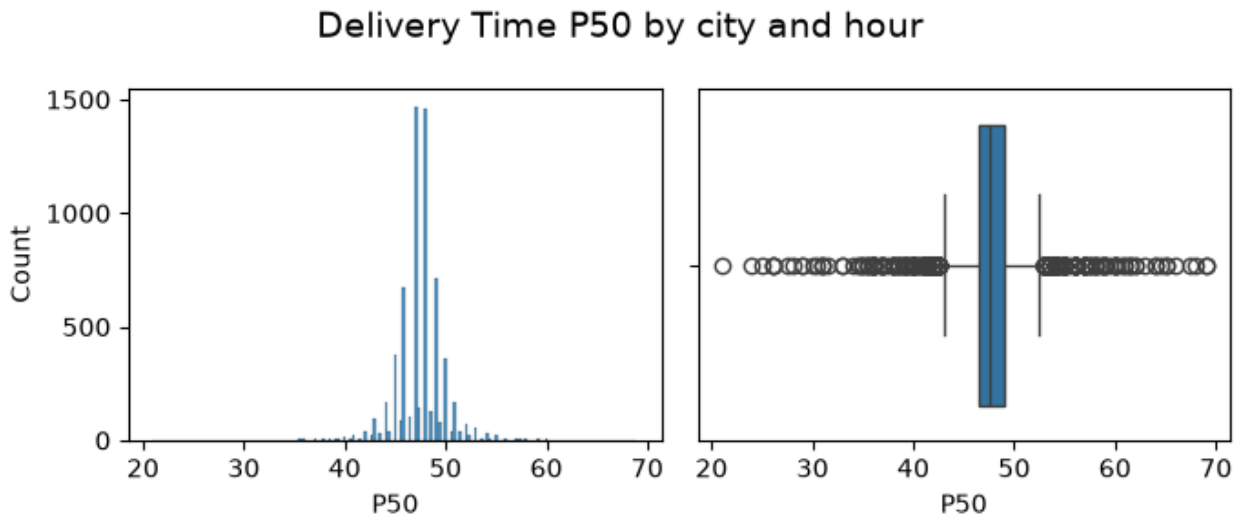
Restaurant rating count distribution

City-day and delivery profile: average order value



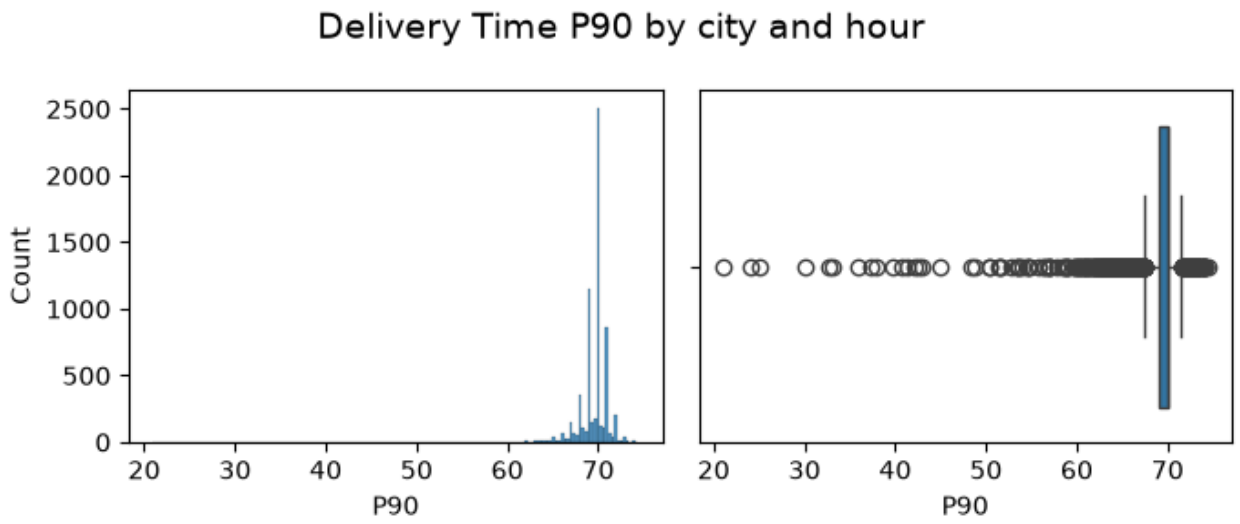
Average order value per city-day

City-day and delivery profile: median delivery time



Median delivery time distribution

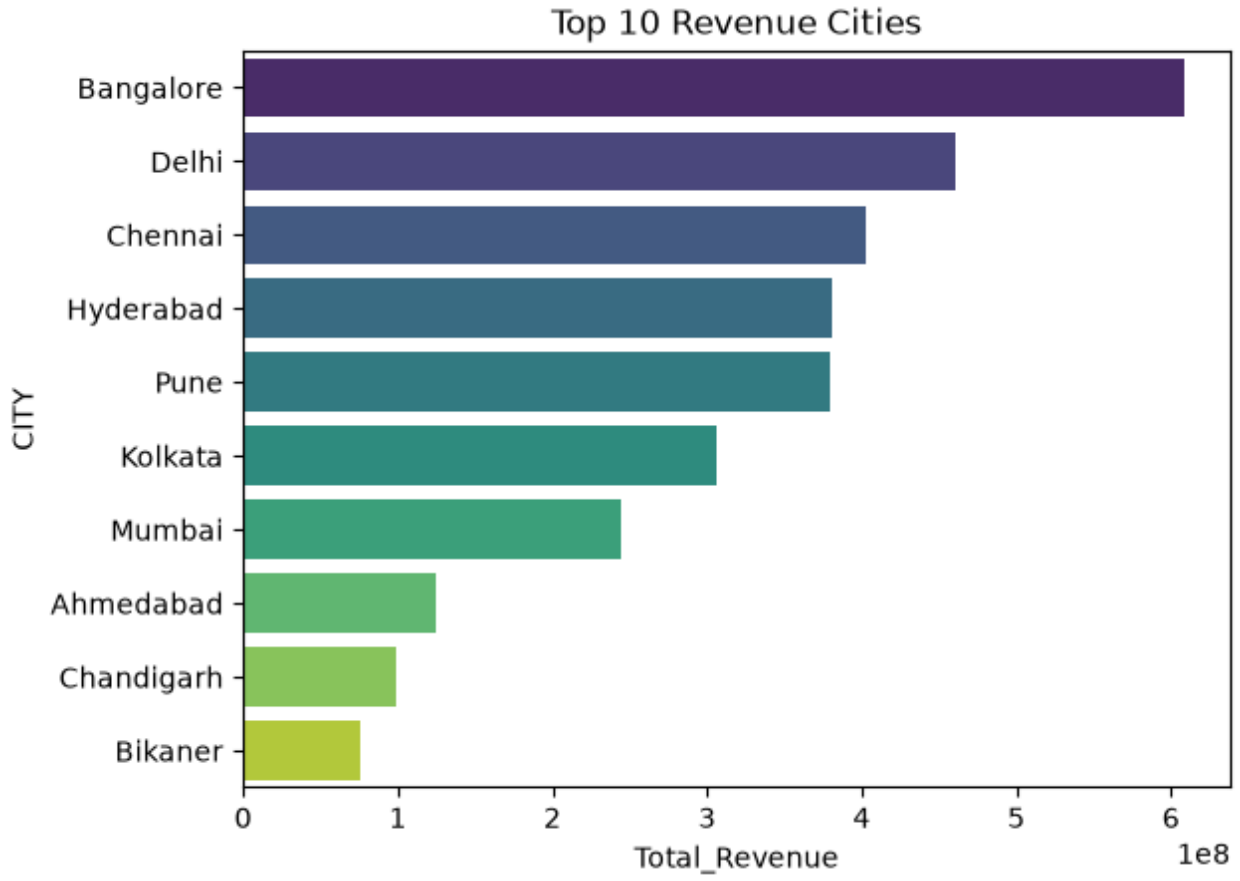
City-day and delivery profile: 90th percentile delivery time



90th percentile delivery time distribution

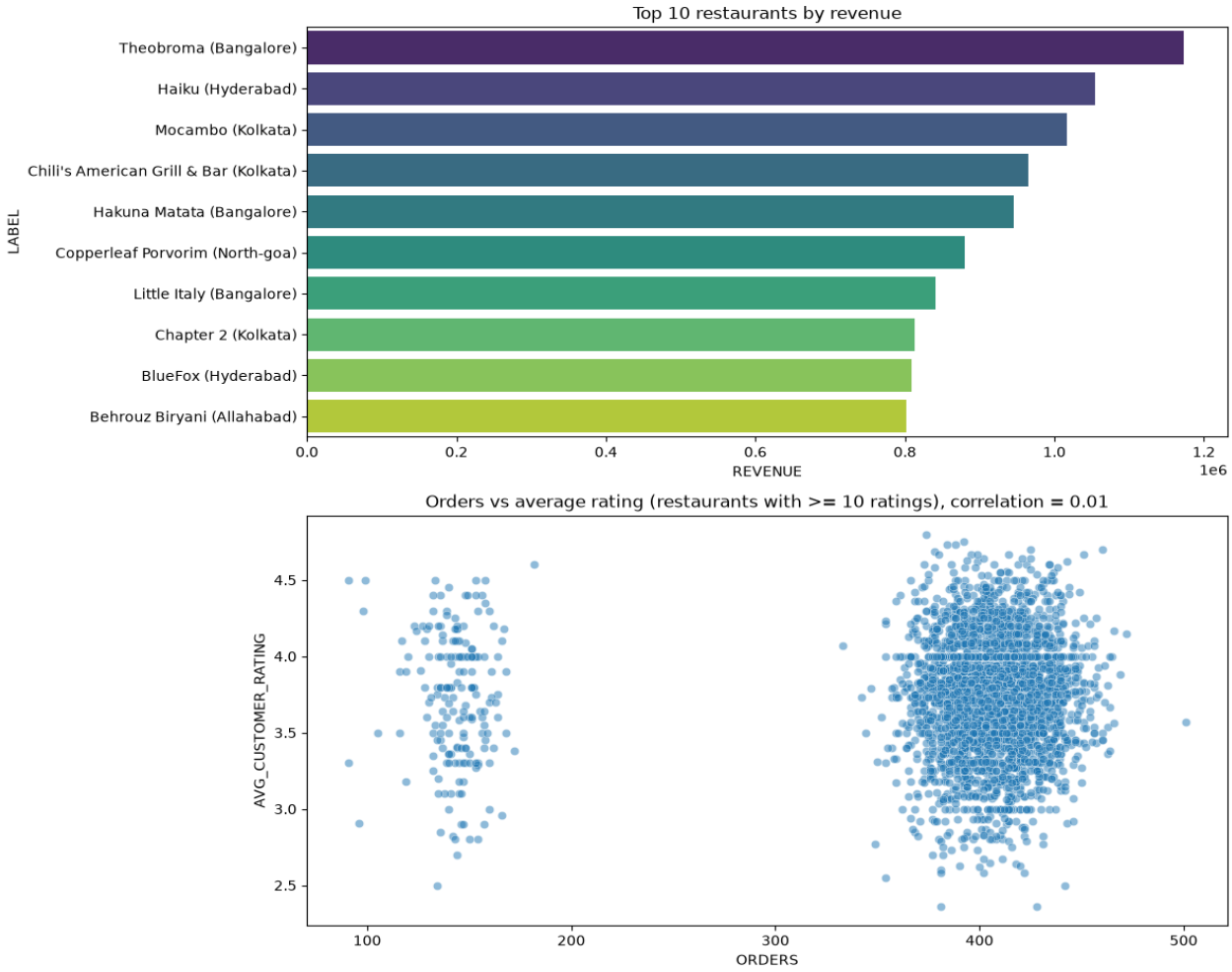
Business and time-series analysis

City revenue and payment mix



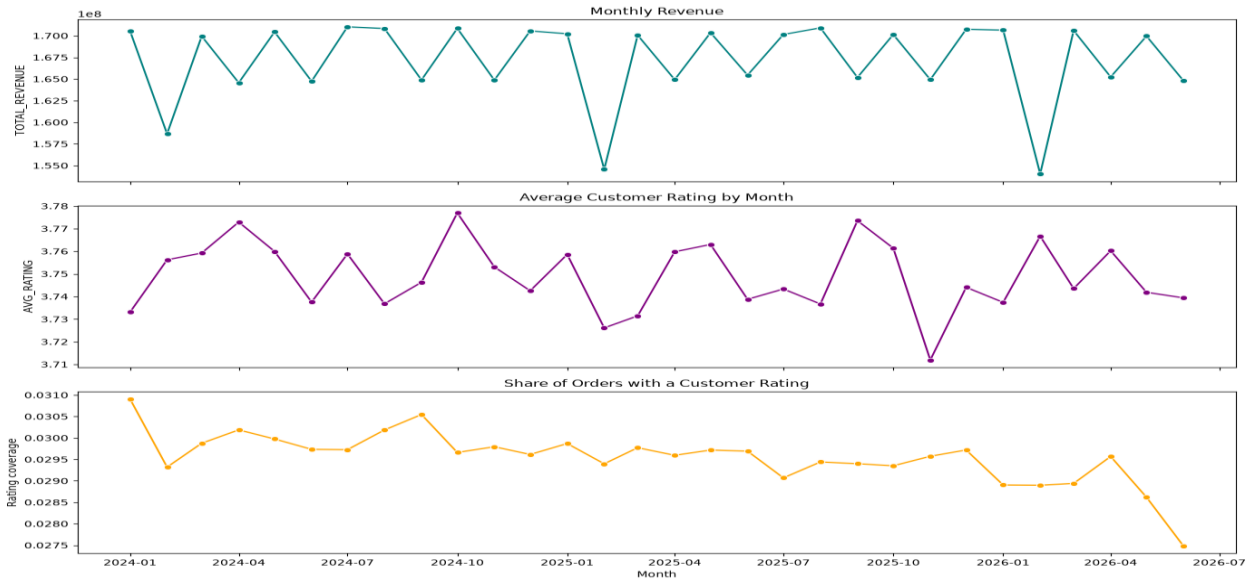
City revenue and payment method summary

Restaurant revenue and rating coverage



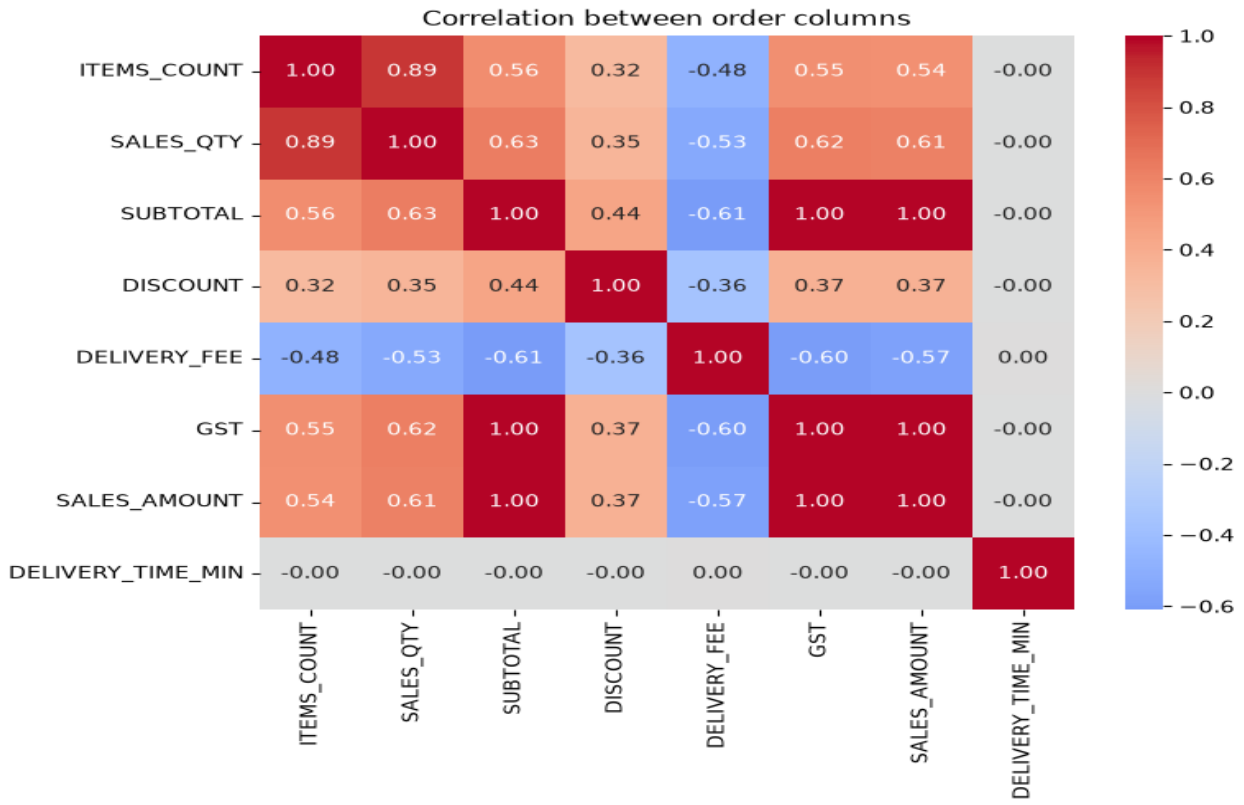
Restaurant revenue and rating coverage

Monthly orders, revenue, ratings, and rating coverage



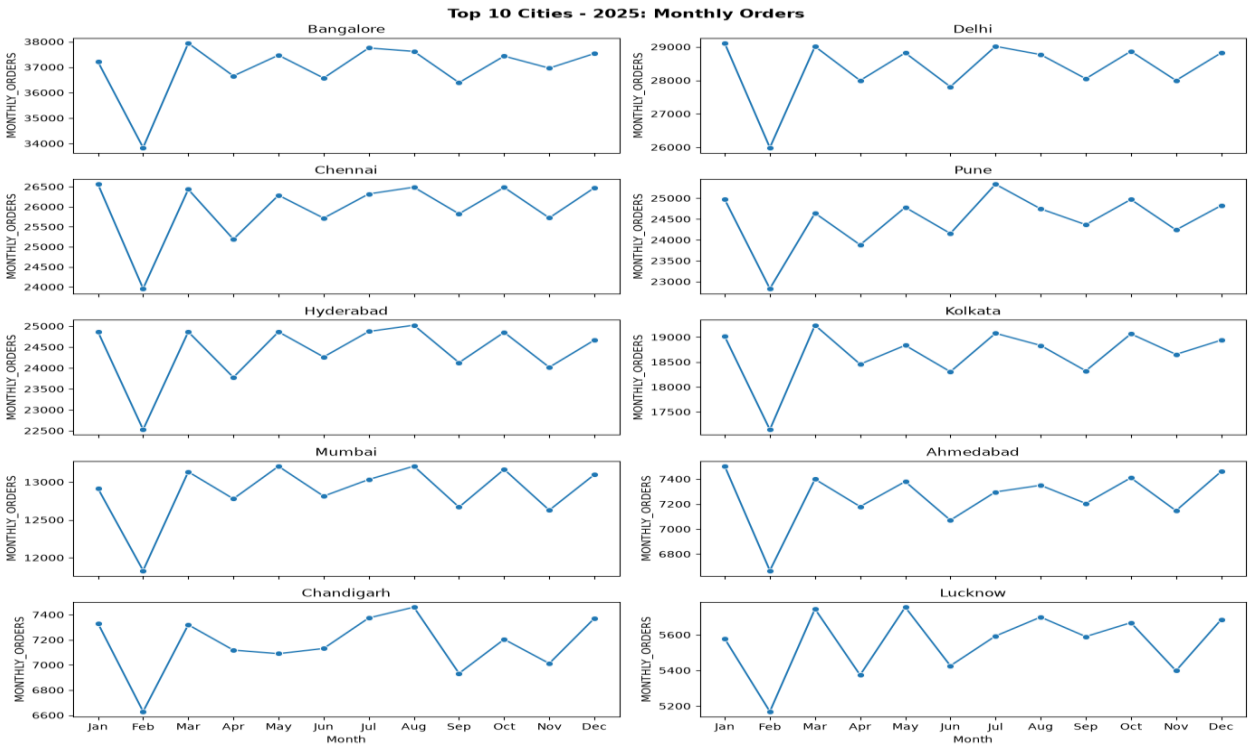
Monthly revenue, ratings, and rating coverage

Order measure correlations



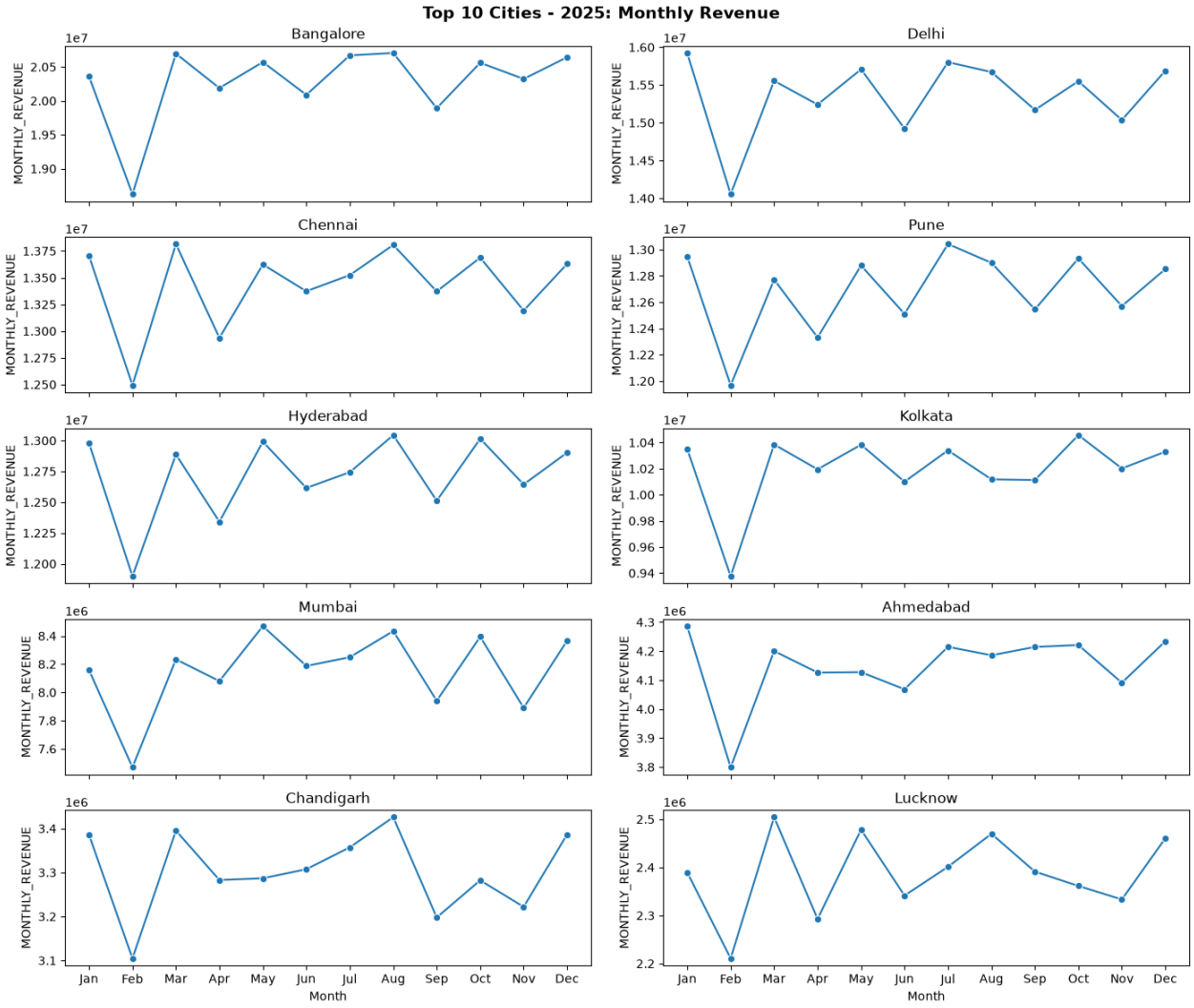
Correlation between order measures

Top cities: monthly order pattern in 2025



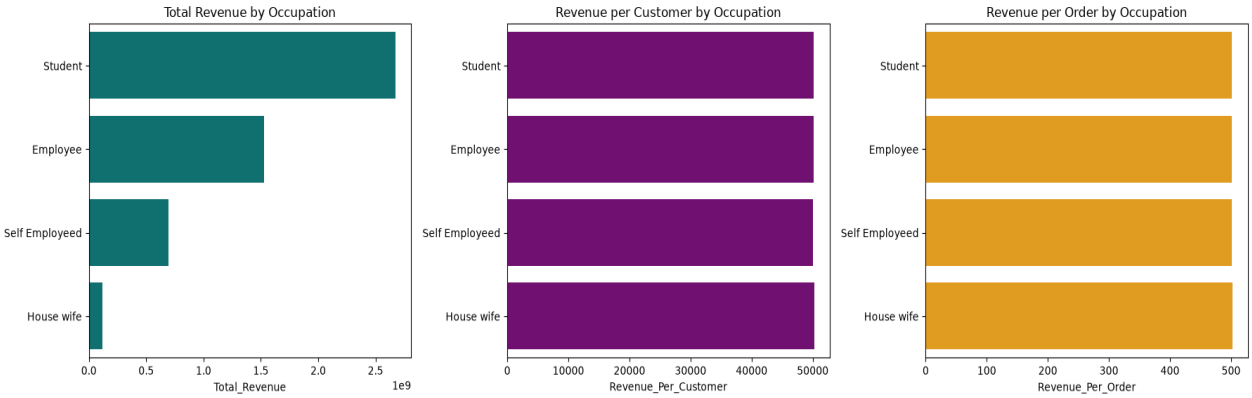
Top cities monthly orders in 2025

Top cities: monthly revenue pattern in 2025



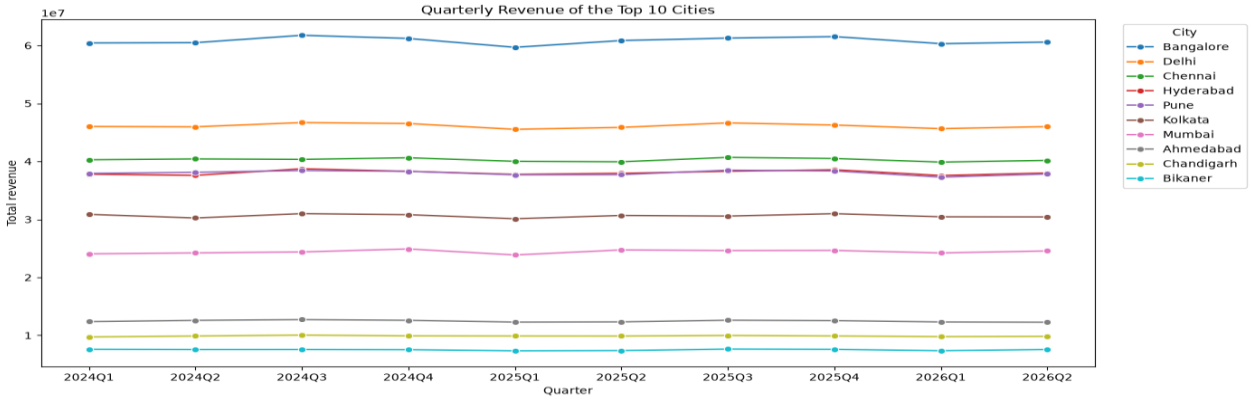
Top cities monthly revenue in 2025

Revenue by customer occupation



Revenue comparison by customer occupation

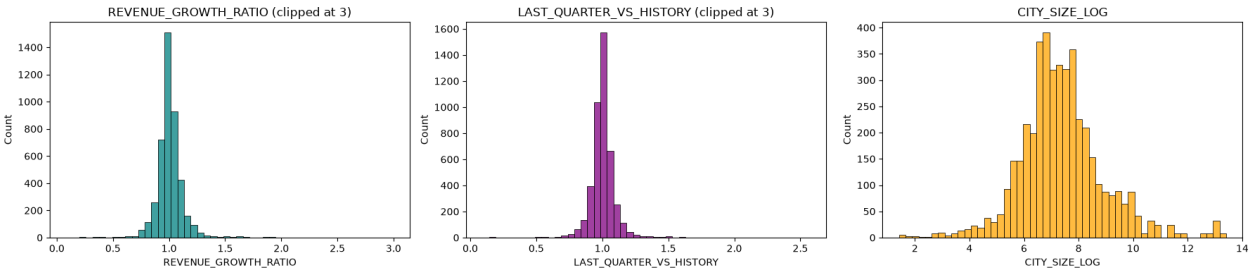
Top-city quarterly revenue pattern



Top cities quarterly revenue

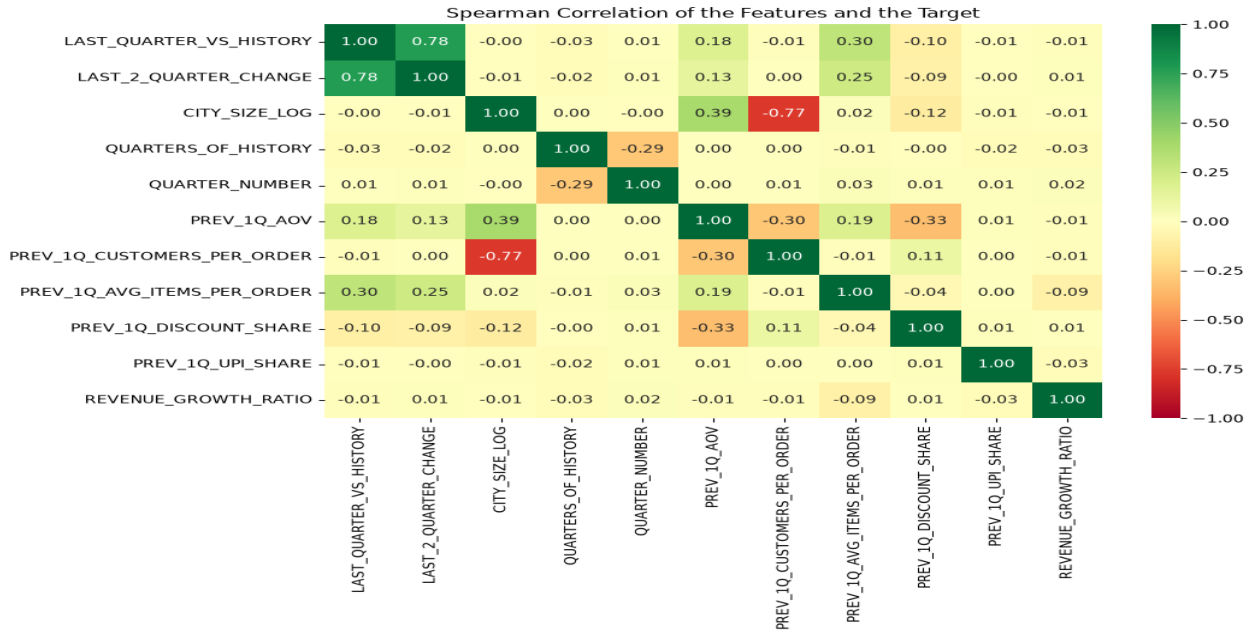
Model diagnostics and evaluation

Engineered-feature checks



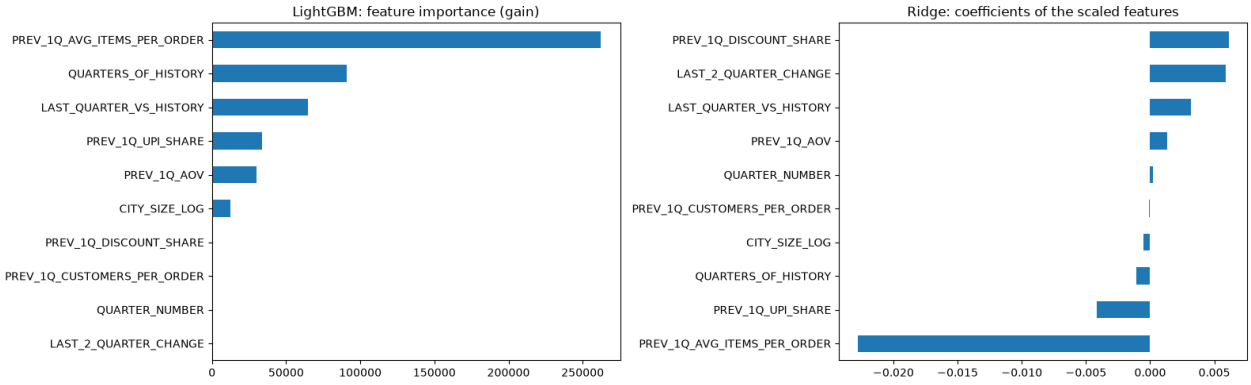
Feature distributions and leakage checks

Feature relationships with target



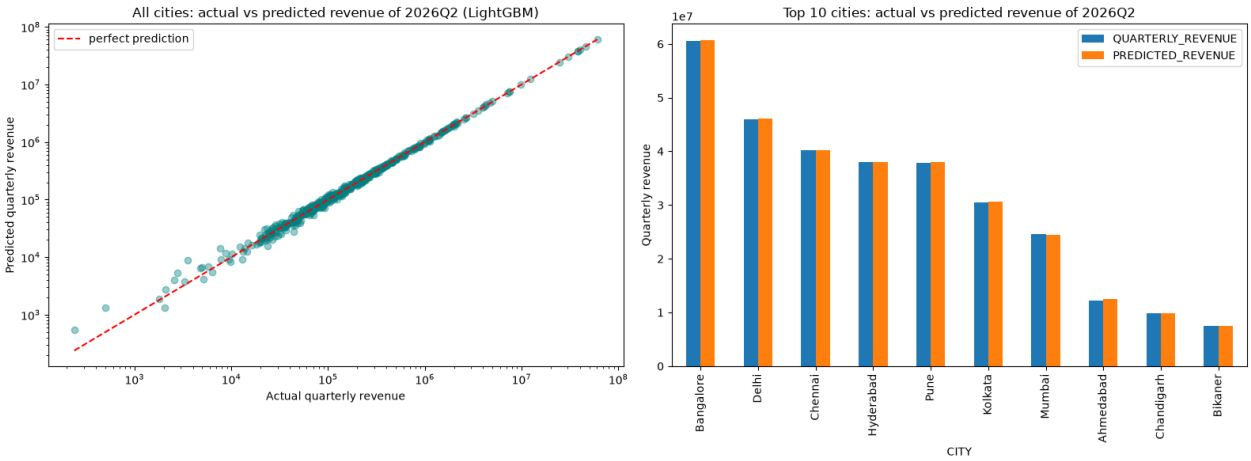
Feature correlation with target

Model feature interpretation



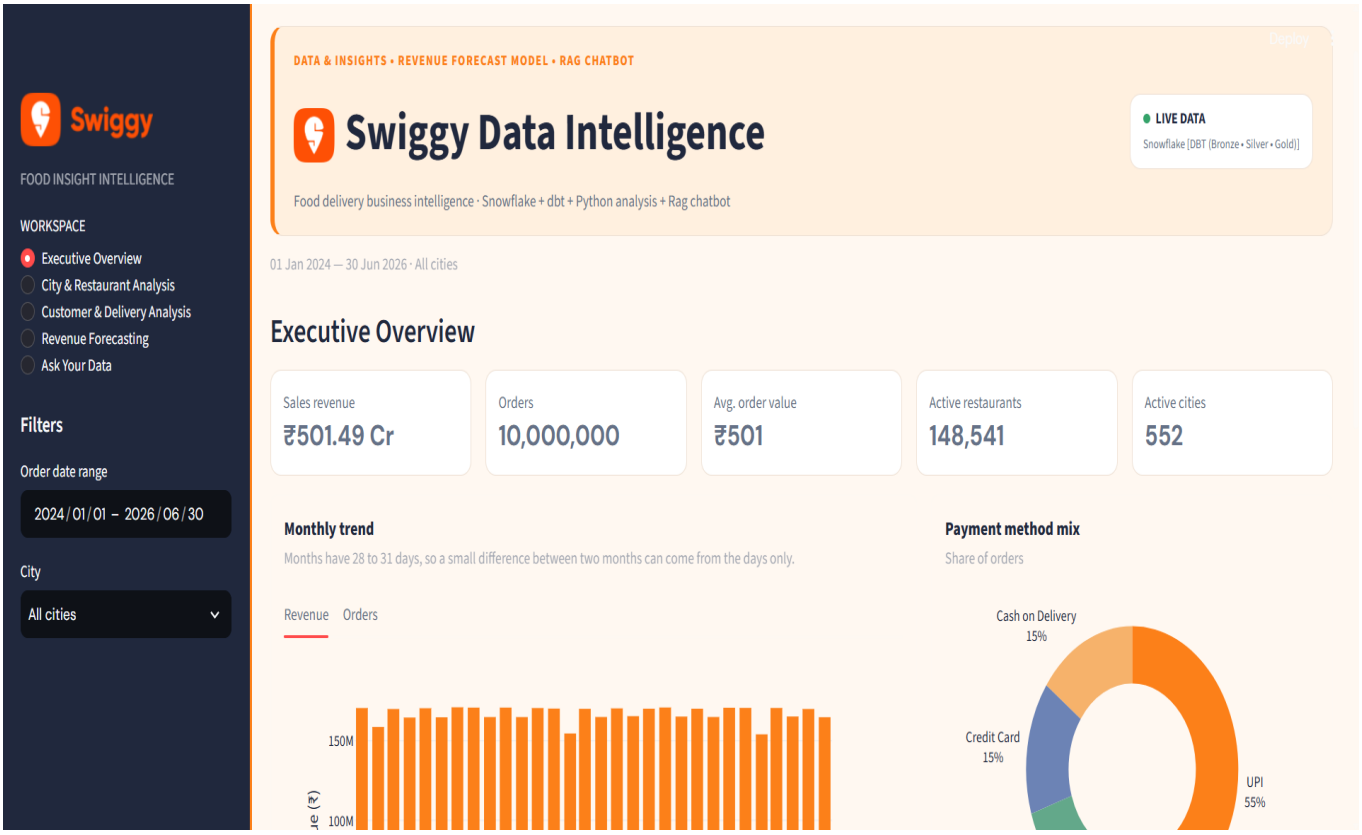
LightGBM feature importance and Ridge coefficients

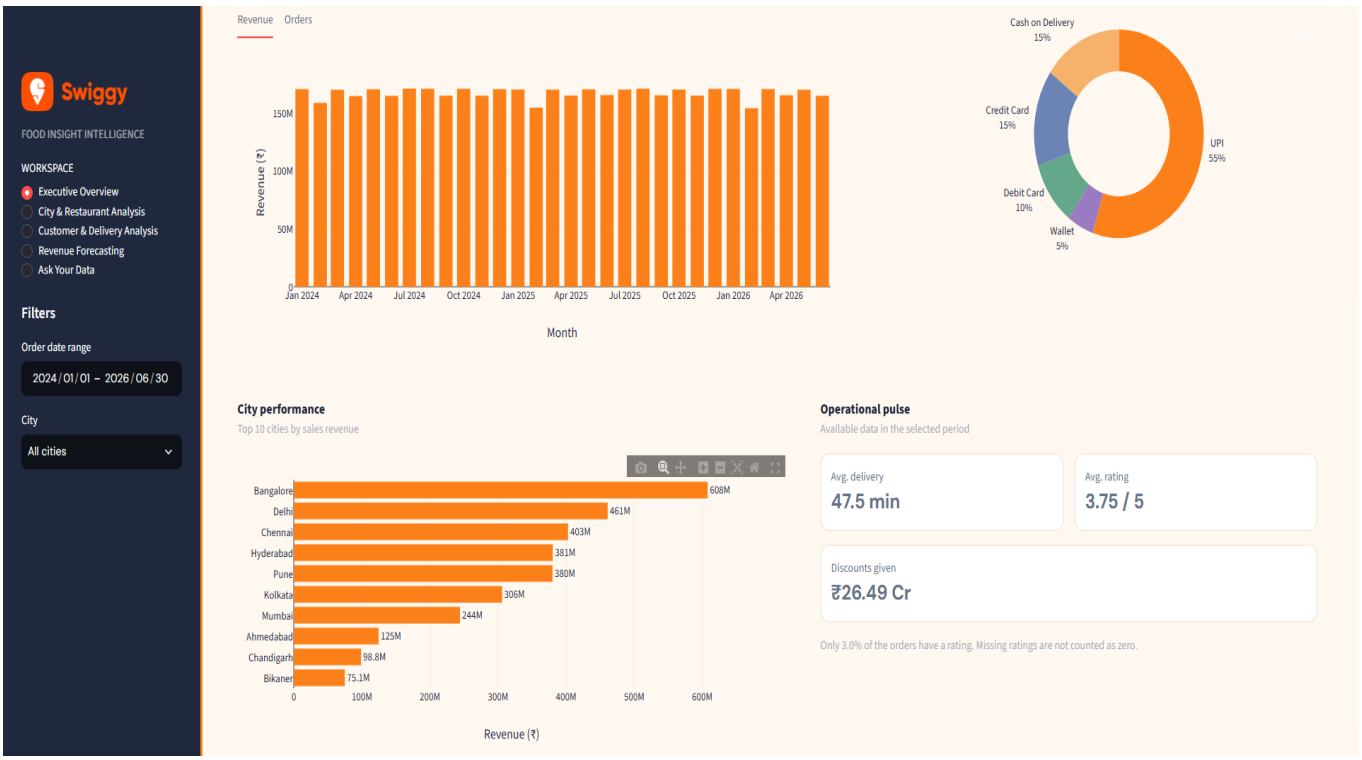
Holdout forecast comparison



Actual versus predicted city revenue for the holdout quarter

Streamlit Dashboard Demo





Swiggy

FOOD INSIGHT INTELLIGENCE

WORKSPACE

- Executive Overview
- City & Restaurant Analysis
- Customer & Delivery Analysis
- Revenue Forecasting
- Ask Your Data

Filters

Order date range

2024/01/01 – 2026/06/30

City

All cities

DATA & INSIGHTS • REVENUE FORECAST MODEL • RAG CHATBOT

Swiggy Data Intelligence

Food delivery business intelligence · Snowflake + dbt + Python analysis + Rag chatbot

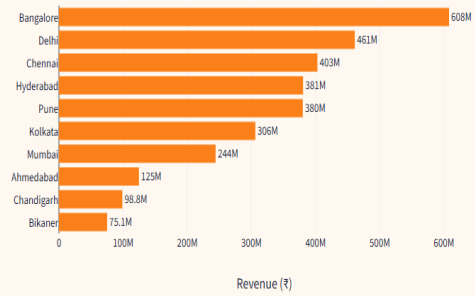
LIVE DATA

Snowflake (DBT (Bronze • Silver • Gold))

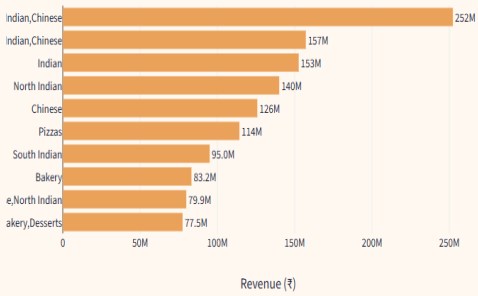
01 Jan 2024 — 30 Jun 2026 · All cities

City & Restaurant Analysis

Top cities



Cuisine performance



CITY	SALES_REVENUE	ORDERS
Bangalore	608065944.78	1106178

CUISINE	ORDERS	REVENUE	AVG_RATING
North Indian,Chinese	465234	252341018.78	3.7504

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All cities

CITY	SALES_REVENUE	ORDERS
Bangalore	608065944.78	1106178
Delhi	461200277.03	851171
Chennai	402913964.61	778888
Hyderabad	380563991.44	731339
Pune	380149131.29	733802
Kolkata	306135600.41	559736
Mumbai	244220181.74	385487
Ahmedabad	124620164.21	217839
Chandigarh	98816702.45	215287
Bikaner	75108042.54	123192

CUISINE	ORDERS	REVENUE	AVG_RATING
North Indian,Chinese	465234	252341018.78	3.7504
Indian,Chinese	291638	157188590.28	3.7245
Indian	341708	152594670.84	3.7561
North Indian	300241	139990873.52	3.7548
Chinese	262710	125820301.97	3.7586
Pizzas	193047	114231814.52	3.7417
South Indian	257854	95008362.05	3.7549
Bakery	147401	83220713.01	3.7476
Chinese,North Indian	156087	79862258.26	3.7427
Bakery,Desserts	147212	77479473.4	3.7039

Top restaurants

RESTAURANT_ID	RESTAURANT_NAME	CITY	CUISINE	ORDERS	REVENUE	AVG_CUSTOMER_RATING	AVG_DELIVERY_MIN
437065	Theobroma	Bangalore	Desserts,Bakery	371	1172959.82	3.9545	47.8329
22964	Haiku	Hyderabad	Japanese,Asian	447	1054275.06	3.9091	47.4586
4484	Mocambo	Kolkata	North Indian,Biryani	405	1016055.43	4.3571	49.3753
16214	Chili's American Grill & Bar	Kolkata	American,Salads	445	965577.76	3.75	46.3258
223614	Hakuna Matata	Bangalore	Chinese,Indian	390	946241.41	4.3636	49.7436
84365	Copperleaf Porvorim	North-goa	Indian,Chinese	449	879774.38	3.65	46.8909
8354	Little Italy	Bangalore	Italian,Pizzas	421	841405.74	3.2857	47.0618
292634	Chapter 2	Kolkata	Continental,European	451	813587.52	3.7222	46.6164
198307	BlueFox	Hyderabad	Biryani,Chinese	423	808388.16	3.5714	47.0969
442050	Behrouz Biryani	Allahabad	Biryani,Mughlai	408	801402.19	4.1111	48.6397

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FOOD INSIGHT INTELLIGENCE

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Filters

Order date range: 2024/01/01 – 2026/06/30

City: All cities

DATA & INSIGHTS • REVENUE FORECAST MODEL • RAG CHATBOT

Swiggy Data Intelligence

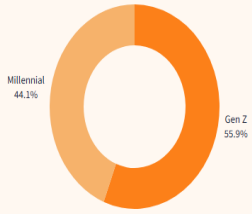
Food delivery business intelligence · Snowflake + dbt + Python analysis + Rag chatbot

LIVE DATA
Snowflake [DBT (Bronze • Silver • Gold)]

01 Jan 2024 — 30 Jun 2026 · All cities

Customer & Delivery Analysis

Customer segments



AGE_SEGMENT	CUSTOMERS	ORDERS	REVENUE
Gen Z	55934	5595154	2805664387.85
Millennial	44066	4404846	2209190992.54

Delivery and rating coverage

Average delivery time	Average customer rating	Rating coverage
47.5 min	3.75 / 5	3.0%

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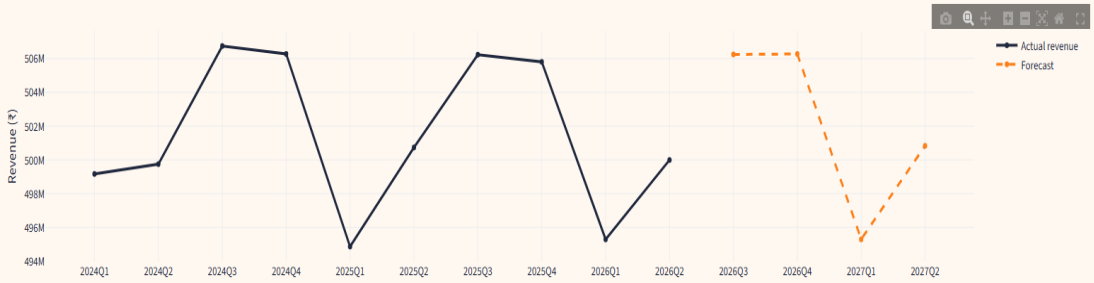
Revenue Forecasting

Past quarters are actual revenue, future quarters are predictions of the saved city_revenue_forecast.joblib model.

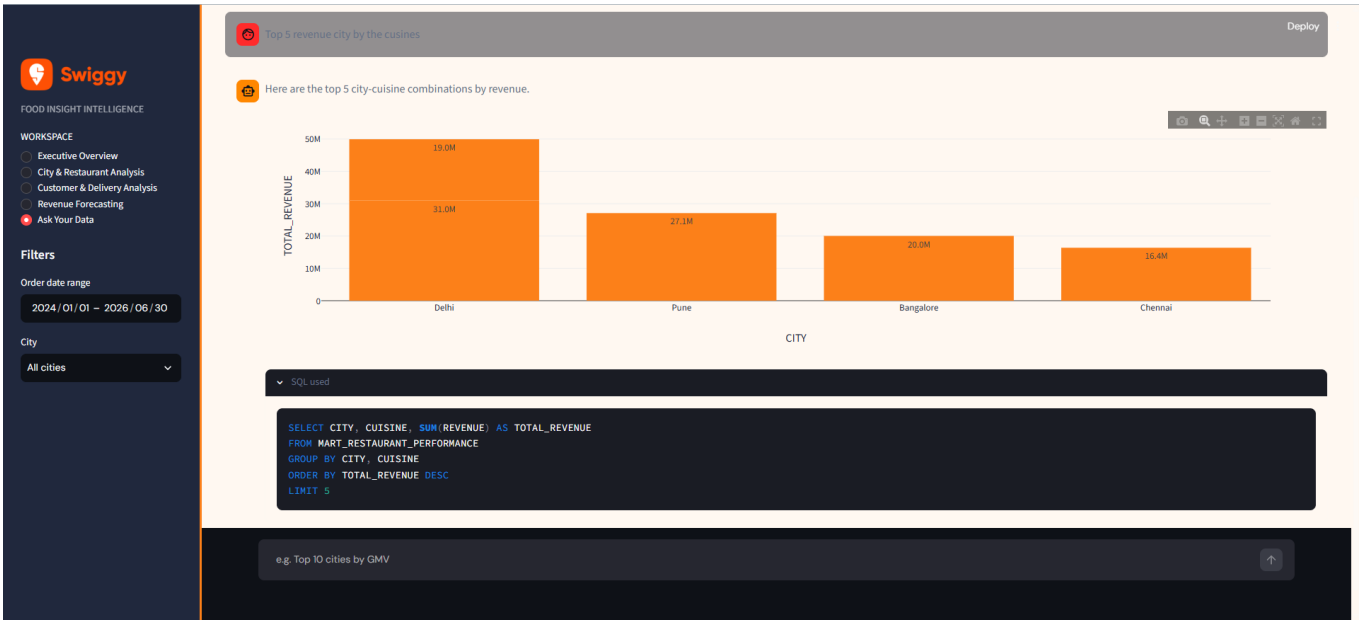
Forecast city: All cities

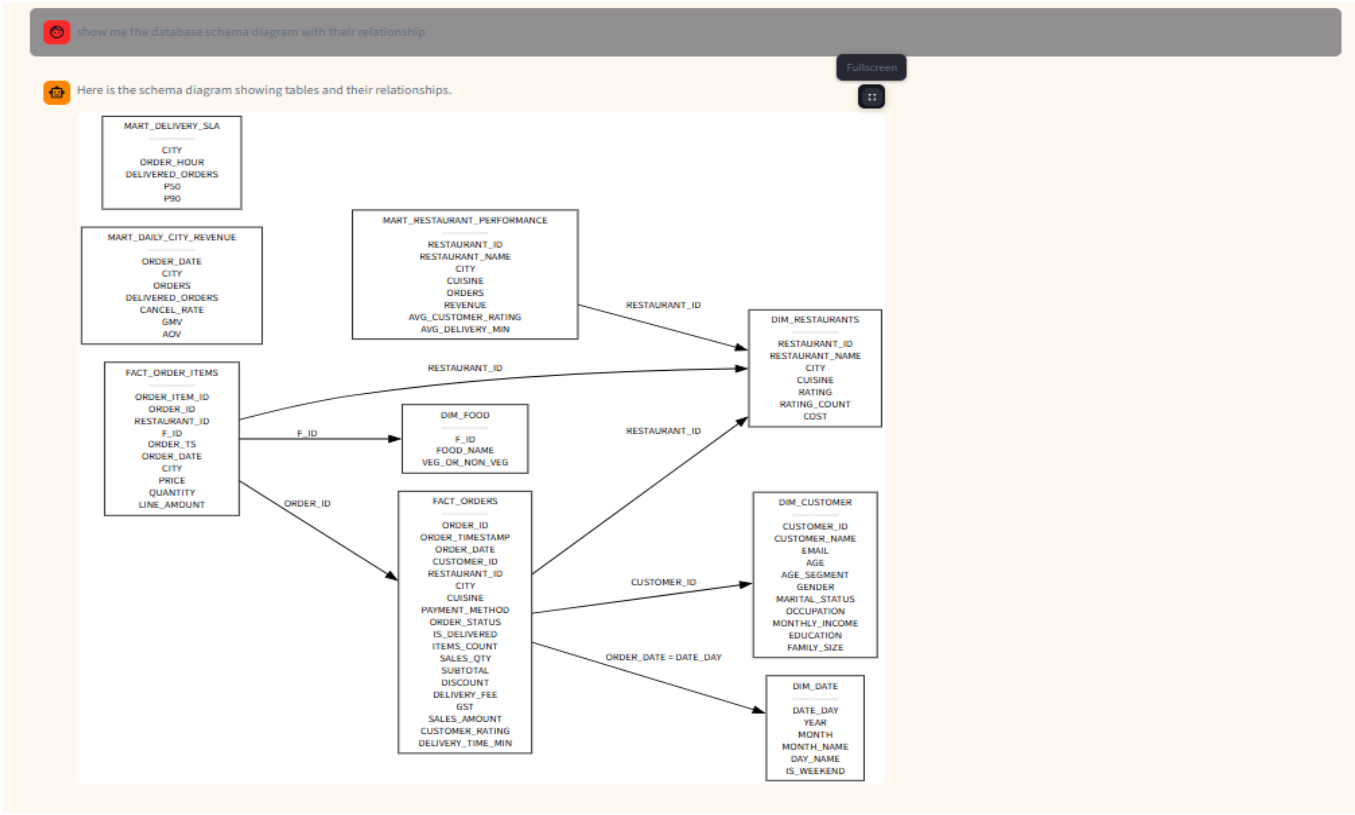
How many quarters to forecast: 4

Next quarter (2026Q3)	Last forecast quarter (2027Q2)
₹50.62 Cr	₹50.08 Cr



CITY	QUARTER	ACTUAL_REVENUE	PREDICTED_REVENUE	PREDICTED_GROWTH_RATIO	ROW_TYPE
All cities	2024Q1	499164602.76	None	None	Actual
All cities	2024Q2	499746898.8	None	None	Actual
All cities	2024Q3	506746397.06	None	None	Actual





Explain in text, the patten related to the customer which type of customer order food most, and which category food cusines, by the top city

To identify the customer segment that orders the most and the most popular cuisine in the top city, you would: 1) Aggregate orders by CUSTOMER_ID (or by demographic fields such as AGE_SEGMENT, GENDER, OCCUPATION, etc.) from FACT_ORDERS, joining DIM_CUSTOMER to get segment details; 2) Rank cities by total order volume (e.g., using MART_DAILY_CITY_REVENUE or summing FACT_ORDERS per CITY) to find the top city; 3) Within that city, group orders by CUISINE (from FACT_ORDERS or DIM_RESTAURANTS) to see which cuisine has the highest order count or revenue; 4) Compare the order counts/revenue across customer segments to see which segment orders the most. The result will show the dominant customer type (e.g., young professionals, families, etc.) and the leading cuisine (e.g., Indian, Chinese) in the city with the highest overall orders.